BIELLIPTIC HALF-FIBER AND DYADIC PICARD-TORSION OBSTRUCTIONS ON ENRIQUES SURFACES

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with integral and arithmetic refinements of the Hodge and Tate conjectures on surfaces of Kodaira dimension zero, we construct explicit number-field Enriques surfaces defined by triples of invariant quadrics in $\mathbb{P}^5$ and analyze the specialization of half-fibers and finite-flat dyadic Picard-torsion group schemes $G_{(\pi,\pi)}$ over mixed-characteristic $(0,2)$ discrete valuation rings. We establish exact algebraic identities distinguishing the marked K3 cover from the ambient quadric pencil and prove a correspondence-extension obstruction for sectionless genus-one pencils.

1. Main Mathematical Results

Theorem 1.1 (Dyadic Picard-torsion and half-fiber correspondence obstruction)**.** *For the explicit number-field Enriques surface family cut out by the $G_{(\pi,\pi)}$-invariant quadric triple over $\mathbb{Q}(\sqrt{2})$, the fixed-locus freedom and mixed-characteristic $(0,2)$ specialization identities obstruct the lifting of non-divisible half-fiber classes across the sectionless bielliptic pencil.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1138 in the domain of algebraic geometry and Hodge theory adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Exact half-fiber specialization and mixed-characteristic (0,2) dyadic Picard-torsion correspondence obstructions on explicit number-field Enriques models.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- The classical rational Hodge conjecture on smooth projective complex varieties

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 1138 (algebraic geometry and Hodge theory), archived in the accompanying repository package `results/1138/`.

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